

CS 486/686

Markov Decision Processes I

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Lecture 13

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Learning goals

- Motivate modeling a decision problem as a **Markov decision process**.
- Describe the **components** of a fully-observable MDP.
- Explain why we use a **discounted** reward function.
- Define the **policy** of an MDP.
- Give examples of how the **reward function** shapes the optimal policy.

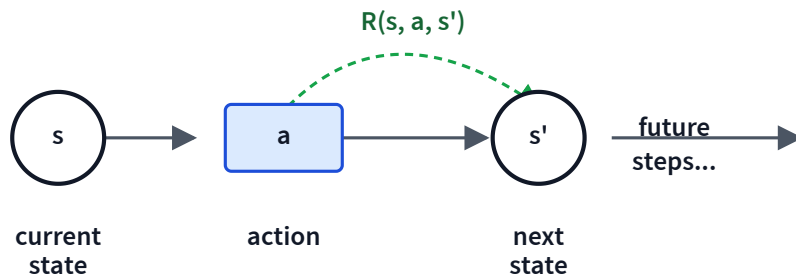
From one-shot decisions to ongoing ones

Two new things

- **Horizon:** the problem may never end.
 - *Infinite horizon:* the process may go on forever.
 - *Indefinite horizon:* the agent will stop eventually, but not at a known time.
- **Rewards along the way**, not just utility at the end.
 - The agent may never reach an "end".
 - Each step's reward includes action costs + bonuses/penalties.

A Markov decision process (MDP) formalizes this.

A Markov decision process



An MDP is a 4-tuple (S, A, P, R) :

- S : set of states.
- A : set of actions.
- $P(s' \mid s, a)$: transition probabilities.
Stationary: same for every t .
- $R(s, a, s')$: reward function.

For this lecture we'll use the simpler form $R(s)$ = reward for entering state s .

How do we aggregate rewards over time?

Total reward

$$\sum_{t=0}^{\infty} R(S_t)$$

May diverge $\Rightarrow$
policies are not
comparable.

Average reward

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=0}^n R(S_t)$$

Zero whenever the total
reward is finite.

Discounted reward

$$\sum_{t=0}^{\infty} \gamma^t R(S_t), \quad \gamma \in [0, 1)$$

Always finite; the standard
choice.

The discounted sum is also called the **return** $G(S_0)$.

The discount factor γ

- **Time preference.** A dollar today is worth more than a dollar tomorrow.
- **Finiteness.** If $R(s) \in [-R_{\max}, R_{\max}]$ and $\gamma \in [0, 1)$:

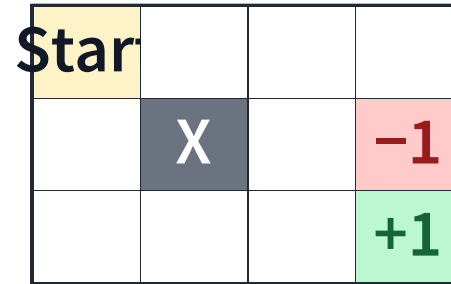
$$|\sum_{t=0}^{\infty} \gamma^t R(S_t)| \leq \frac{R_{\max}}{1 - \gamma}.$$
- **Effective horizon.** Roughly $\frac{1}{1-\gamma}$ steps matter (e.g. $\gamma = 0.9 \Rightarrow$ about 10 steps).

Variants

- **Fully-observable MDP** — the agent always knows its current state. (Today.)
- **Partially observable MDP (POMDP)** — combine MDP + HMM; the state is hidden behind observations.

A 3×4 grid world

What should the robot do to maximize its rewards?



- Let s_{ij} be row i , column j . s_{11} is the start.
- s_{22} is a wall. s_{24} and s_{34} are goal states — the robot exits the world there.

The grid world as an MDP

Star			
	X		-1
			+1

Actions: up / down / left / right (in every state).

Transition $P(s' \mid s, a)$:

- 0.8 — action takes intended effect.
- 0.1 — 90° left of intended.
- 0.1 — 90° right of intended.
- If the robot bumps a wall, it stays put.

Reward $R(s)$: $R(s_{24}) = -1$, $R(s_{34}) = +1$; otherwise $R(s) = -0.04$.

Understanding the transition model

Q1. The robot is in s_{14} (top-right corner) and tries to move right. What is the probability it *stays* in s_{14} ?

Star			s_{14}
	X		-1
			+1

- A. 0.1
- B. 0.2
- C. 0.8
- D. 0.9
- E. 1.0

D — 0.9. Trying right: 0.8 it bumps the wall (stays) + 0.1 it veers up (also bumps wall, stays) = 0.9. The remaining 0.1 veers down to s_{24} .

Why a fixed action sequence is not enough

Q2. In a *deterministic* env, the sequence *down, down, right, right, right* is optimal for the grid world.

Star			
	X		-1
			+1

True. Five steps take us from s_{11} to s_{34} .

Q3. In the *stochastic* env, the same sequence may end at *more than one* square.

True. E.g. five right-veers in a row send us to s_{12} .

Q4. So in the stochastic env, a fixed sequence is still good enough.

False. Actions misfire — we need to *re-plan* from whatever state we end up in.

Need an action for **every state** — a **policy**.

A policy

Policy

A **policy** π specifies the agent's action as a function of the current state.

- **Stationary** — depends only on the state: $\pi : \mathcal{S} \rightarrow \mathcal{A}$.
- **Non-stationary** — also depends on the time step: $\pi : \mathcal{S} \times \mathbb{N} \rightarrow \mathcal{A}$.

For an infinite-horizon fully-observable MDP, the optimal policy is **stationary**.

The reward function reshapes the optimal policy

Same dynamics; different $R(s)$ for non-goal states — very different optimal behaviors.

We will look at five qualitatively different reward regions:

1. $R(s) < -1.6284$ — *life is painful*
2. $-0.4278 < R(s) < -0.0850$ — *life is quite unpleasant*
3. $R(s) = -0.04$ — *life is unpleasant* (the default)
4. $-0.0221 < R(s) \leq 0$ — *life is only slightly dreary*
5. $R(s) > 0$ — *life is GOOD*

Each region balances **risk** (falling into -1) against **reward** (reaching $+1$ quickly).

Five reward regions, five optimal policies

painful

$$R < -1.6284$$

→	→	→	↓
↓	X	→	-1
→	→	→	+1

head to nearest
exit, even -1

quite unpleasant

$$-0.43 < R < -0.085$$

↓	→	↓	←
↓	X	↓	-1
→	→	→	+1

shortcut through
 s_{13}

unpleasant

$$R = -0.04$$

↓	←	←	←
↓	X	↓	-1
→	→	→	+1

long way around;
avoid -1

slightly dreary

$$-0.022 < R \leq 0$$

↓	←	←	↑
↓	X	←	-1
→	→	→	+1

no risk; bang
against walls

GOOD

$$R > 0$$

⊕	⊕	⊕	↑
⊕	X	←	-1
⊕	⊕	←	+1

wander forever;
avoid both goals

Watch the action at s_{13} (top, third column): it changes with every region.

The value function

Expected utility of a policy

- $V^\pi(s)$: expected utility of entering state s and following π thereafter.
- $V^*(s)$: expected utility of entering s and following the **optimal** policy π^* .

0.70	0.60	0.60	0.388
0.76	X	0.66	-1
0.80	0.80	0.91	+1

$V^*(s)$ for $\gamma = 1$, $R(s) = -0.04$ on non-goal states.

The Q-function and the optimal policy

Q-function

$$Q^*(s, a) = R(s) + \sum_{s'} P(s' \mid s, a) V^*(s')$$

"If I were in s and took action a now, then followed π^* forever after — how good is my future?"

$$\pi^*(s) = \arg \max_a Q^*(s, a)$$

Given V^* , the optimal policy is one greedy step away.

Optimal action at s_{13} ?

Q5. Using $Q^*(s, a) = R(s) + \sum_{s'} P(s' | s, a) V^*(s')$, what is $\pi^*(s_{13})$?

0	.70	.60	.61	.388
0	.76	X	.66	-1
0	.80	.80	.91	+1

s_{13} is the yellow cell ($V^* = 0.611$).

Computing each Q-value (4 actions):

Left: $-0.04 + 0.8 \cdot 0.655 + 0.1 \cdot 0.611 + 0.1 \cdot 0.660 = 0.6111$

Right: $-0.04 + 0.8 \cdot 0.388 + 0.1 \cdot 0.611 + 0.1 \cdot 0.660 = 0.3975$

Down: $-0.04 + 0.8 \cdot 0.660 + 0.1 \cdot 0.388 + 0.1 \cdot 0.655 = 0.5923$ Up: $-0.04 + 0.8 \cdot 0.611 + 0.1 \cdot 0.655 + 0.1 \cdot 0.388 = 0.5531$

C — Left. Left has the highest Q^* (0.6111). It moves

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Next: how do we compute V^* ?

Today: given V^* , the optimal policy is just an argmax. But how do we *find* V^* in the first place?

L14: Bellman equations & value iteration.